

Oriented Area-Regge Asymptotics of a Native Four-Simplex Vertex

A single geometric saddle on nondegenerate coherent data

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Abstract

We study an orientation-resolved Riemannian four-simplex vertex, W2, built from a finite relational spin model. Orientation is fixed by restricting the history integral to the domain on which a native $\text{Spin}(4)$ -invariant pseudoscalar Ω has fixed sign. The native positive volume insertion is $|Q|^{1/2}$ on each tetrahedron. No per-face sign variable, Barrett–Crane projection, or additional Einstein–Hilbert boundary projector enters the vertex analysed here.

For boundary data given by Livine–Speziale coherent intertwiners of a nondegenerate four-simplex, with spins $j_f = \lambda k_f + O(1)$, we show, subject to the one hypothesis recorded below: (i) the integrand is exponentially small except near the four pairings of the two chiral gluing solutions; (ii) exactly one of these, a geometric saddle, lies in the interior of the orientation domain, and every configuration on the domain boundary is either exponentially suppressed or rank one, where the measure vanishes cubically and the contribution is subleading; (iii) the wrong-sign volume weight of the boundary state is $O(\lambda^{-\infty})$, by an elementary ladder-operator estimate. The real part of the coherent-action Hessian is proved exactly to be negative definite modulo gauge on both the regular and the Dittrich–Kogios simplex; this also makes the full complex gauge-fixed Hessian nonsingular. The vertex is therefore asymptotic to a single oriented exponential $e^{i\sigma^2 \sum_f j_f \Theta_f}$, the oriented area-Regge phase.

We do *not* obtain a continuum statement. An earlier version inferred, from gluing, the linearised graviton dynamics that Dittrich and Kogios derive from the area-Regge action. That inference is withdrawn: contracting coherent intertwiners across a shared tetrahedron forces the two shapes to agree, whereas the Dittrich–Kogios configuration space relies on the shape mismatch such gluing suppresses (Section 11).

The result is Riemannian and leading-order, holds for coherent boundary data of nondegenerate simplices, and is a vertex statement: it does not control the bulk spin sum. For the model’s own selected tetrahedral state the corresponding sign-concentration statement is reduced exactly to a spectral conjecture that we verify for $j \leq 20$ but do not prove.

No unproved Hessian hypothesis remains at the stated regular/DK scope. The real-part quadratic form is an explicit weighted sum of squares, and exact bivector determinants show that its kernel is precisely the common $SU(2)$ gauge in each chirality. Negative definiteness of the real part then implies invertibility of the complex gauge-fixed Hessian. The uniform norm asymptotic that an earlier version assumed is likewise proved as Lemma 6.2.

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1 Introduction

The claim, and its boundary. We show that one specific four-simplex vertex, derived inside a finite relational spin model, is dominated at large spin by a single nondegenerate geometric saddle whose phase is the oriented area-Regge action. It is natural to hope that this places the vertex at the start of the chain

$$\mathbf{W2} \longrightarrow \text{oriented area Regge} \longrightarrow \text{linearised Riemannian graviton dynamics},$$

whose second arrow is a theorem of Dittrich and Kogios [5]. **We do not establish the first arrow:** gluing coherent vertices forces matched tetrahedral shapes, which is precisely the freedom the Dittrich–Kogios configuration space needs (Section 11). We do not claim to derive general relativity. The statement is Riemannian, semiclassical, leading-order and linearised; it holds for coherent boundary data of nondegenerate simplices; it controls each vertex but not the bulk spin sum of glued histories; and it says nothing about Lorentzian signature or nonlinear curvature. These limits are collected in Section 13.

What is standard. Most of the chain is not new, and we say so where it appears: coherent intertwiners and their norm asymptotics [9]; the two parity-related solutions of the chiral gluing equations for nondegenerate Regge-like data [1]; the Schläfli identity [15]; the use of a

projector to select the Einstein–Hilbert sector in Engle’s proper vertex [7]; and the area-Regge refinement theorem [5].

Two further ingredients are standard and we claim nothing for them. The algebra generated by the pair operators and Q on a four-valent intertwiner is a realisation of the *Racah algebra* $R(3)$, whose generators are the intermediate Casimirs and which is precisely the commutant of the diagonal action [10]; the factorisation of Proposition 2.1 is a one-line consequence of its relations, and the algebraic structure of the four-valent volume operator has been studied at length [13, 14]. Likewise the moment bound of Lemma 6.1 is ordinary spin-coherent-state estimation; its only virtue is that it replaces a symbol calculus at the spectral cusp.

What is specific to this construction. (a) Orientation is selected by a domain restriction on the integration variables, through a pseudoscalar Ω built from the native history variables, rather than by a projector on the boundary state or a per-face sign. (b) The volume insertions and the cofactor $|\Omega_c|$ give the native W2 weight. Its geometric cofactor identity and cubic spacetime-volume power are exact, with no fitted exponent; an exact foliation-free transcription of the full continuum measure of [16] is not needed for the theorem and is not claimed here. We have not located (a) or (b) in the literature; that is a search result, not a novelty theorem, and the programme that produced this paper has repeatedly found native-looking structures to be standard. The contribution we do claim is the assembly: a single-saddle asymptotic for this vertex in which boundary control and sign concentration are proved rather than assumed.

On method. Every load-bearing numerical statement was frozen with its log, reproduced by an implementation sharing no code, and submitted to hostile review by reviewers from more than one model provider, including context-free sessions. Every search for the absence of something carries a positive control that must first find a known instance; a search whose control fails issues no verdict. The review trail, including the findings that changed this text, is in Appendix A.

AI collaboration. The derivations, code, numerical experiments and drafting of this paper were carried out in collaboration with large language models, from more than one provider, working under the direction of the author. The author set the questions, decided which lines to pursue, adjudicated the results, and is responsible for the content; no model is or could be an author, since authorship carries accountability for the claims and that rests with a person.

We state this because the method is load-bearing rather than incidental. Every load-bearing numerical claim was reproduced by an independent implementation sharing no code path; every search for the absence of something carried a positive control that had to recover a known instance before any verdict was issued; and each stage was submitted to adversarial review by models from different providers, instructed to falsify rather than to improve. Several results in this paper exist in their present form only because such a review falsified an earlier version, and those corrections are recorded rather than quietly absorbed.

2 The native selector

On the intertwiner space $\mathcal{K}$ of a four-valent node let A, B, C be the three pair operators, trace-centred to a, b, c with $a + b + c = 0$, and put

$$Q = i[a, b] = \mathbf{J}_1 \cdot (\mathbf{J}_2 \times \mathbf{J}_3), \quad G = a^2 + b^2 + c^2.$$

These generate a realisation of the Racah algebra $R(3)$: its generators are the intermediate Casimirs $C^{(12)}, C^{(23)}$, which are the pair operators up to a shift, and the algebra is the

commutant of the diagonal action [10]. The following identity is an elementary consequence of the two standard relations $a + b + c = 0$ and $[a, b] = [b, c] = [c, a]$. We record it because it is what makes the selector positive and its ground state sharply defined, not because it is new.

Proposition 2.1 (Exact factorisation). *With $\omega = e^{2\pi i/3}$ and $Z = a + \omega b + \omega^2 c$,*

$$ZZ^\dagger = \frac{3}{2}(G - \sqrt{3}Q), \quad Z^\dagger Z = \frac{3}{2}(G + \sqrt{3}Q).$$

Hence $H_\sigma := G - \sigma\sqrt{3}Q \geq 0$, $H_+ = \frac{2}{3}ZZ^\dagger$, $H_- = \frac{2}{3}Z^\dagger Z$, and

$$Q = \frac{Z^\dagger Z - ZZ^\dagger}{3\sqrt{3}}. \quad (2.1)$$

Proof. Direct expansion, using $\omega + \omega^2 = -1$, $\omega - \omega^2 = i\sqrt{3}$, $(a + b + c)^2 = 0$ (which gives $\{a, b\} + \{b, c\} + \{c, a\} = -G$), and $[a, b] = [b, c] = [c, a] = -iQ$ (a consequence of $a + b + c = 0$). The symmetric part of $ZZ^\dagger$ is $G + \frac{1}{2}G$ and the antisymmetric part is $-\frac{3}{2}\sqrt{3}Q$; reversing the order flips the sign of the latter. $\square$

The identity holds for arbitrary external spins. Numerically it was verified for equal spins $j = \frac{1}{2}, \dots, 6$ (residuals $\leq 2.1 \times 10^{-12}$), and (2.1) for $j \leq 20$ (residuals $\leq 3.6 \times 10^{-15}$).

The ground state of H_+ is the model's own tetrahedral state. It enters only in Section 12, which treats it separately from the main theorem.

3 The oriented vertex W2

Let $X_a^\pm \in SU(2)$, $a = 0, \dots, 4$, be the doubled history variables and $N_a = X_a^-(X_a^+)^{-1}$, read as unit vectors in $\mathbb{R}^4$.

Definition 3.1 (Vertex). Put $\Omega_a = (-1)^a \det(N_0, \dots, \widehat{N}_a, \dots, N_4)$ and $\mathcal{D}_\sigma = \{X : \text{sgn } \Omega_a(X) = \sigma \forall a\}$. For a boundary state Ψ on $\bigotimes_e \mathcal{K}_e^+ \otimes \mathcal{K}_e^-$,

$$\mathcal{A}_\sigma[\Psi] = \int_{\mathcal{D}_\sigma} dX |\Omega_c(X)| \mathcal{I}_X[(\bigotimes_e \widehat{T}_e)\Psi], \quad \widehat{T}_e \propto |Q_e|^{1/2}.$$

where $\mathcal{I}_X$ is the ordinary doubled K_5 matrix element with $U_{ab}^\pm = D^{jab}((X_a^\pm)^{-1}X_b^\pm)$.

The following are proved in [17]: Ω_a , $\mathcal{D}_\sigma$ and $|\Omega_c|$ are $\text{Spin}(4)$ invariant; $\mathcal{A}_\sigma$ is linear in Ψ ; and the exchange $X^+ \leftrightarrow X^-$ maps $\mathcal{D}_+ \leftrightarrow \mathcal{D}_-$. The native volume/cofactor weight is fixed by the model. Its geometric core can be proved directly: if T_a is the volume of the facet opposite vertex a , V_4 the four-volume and λ_a the barycentric coordinate, then $\nabla \lambda_a = -(T_a/4V_4)N_a$ and $|\det(\nabla \lambda_b)_{b \neq a}| = (4!V_4)^{-1}$. Hence

$$|\Omega_a| = \frac{32}{3} \frac{V_4^3 T_a}{\prod_b T_b}.$$

Together with $T_e^2 = (2/9)|\mathbf{E}_1 \cdot (\mathbf{E}_2 \times \mathbf{E}_3)|$ and $i[\mathbf{J}_1 \cdot \mathbf{J}_2, \mathbf{J}_2 \cdot \mathbf{J}_3] = \mathbf{J}_1 \cdot (\mathbf{J}_2 \times \mathbf{J}_3)$, this gives the exact native cubic-volume identity recorded in the companion. The stronger claim that this is a unique foliation-free discretisation of the full continuum measure is not used below.

4 Coherent boundary data and the doubled action

Standing assumption 4.1. Fix a nondegenerate Riemannian four-simplex with face areas $k_f > 0$ and outward face normals $n_{ab} \in S^2$ in each tetrahedral frame. Let $\lambda \rightarrow \infty$ along a sequence with $j_f = \lambda k_f + O(1)$ admissible (half-integral and satisfying the coupling conditions). For each tetrahedron let $\Phi_e = \bigotimes_{f \ni e} |j_f, n_{ef}\rangle$ and $\iota_e = \Pi \Phi_e / \|\Pi \Phi_e\|$, $\Pi = \int_{SU(2)} dg U(g)$, be the Livine–Speziale coherent intertwiner [9]. The boundary state is $\Psi = \bigotimes_e \iota_e \otimes \bar{\iota}_e$.

For the Dittrich–Kogios (DK) simplex we take the vertices

$$(0, 0, 0, 0), (1, 0, 0, 0), (1, 1, 0, 0), (1, 1, 1, 0), \frac{1}{2}(1, 1, 1, 1),$$

whose ten face areas stand in the ratios $\frac{\sqrt{3}}{4} : \frac{1}{2} : \frac{\sqrt{2}}{2}$ with multiplicities $4 : 4 : 2$. We identify the simplex by its coordinates, not by its areas, since ten areas do not in general determine a four-simplex. The area ratios are irrational, which is why the spin sequence is $\lambda k_f + O(1)$ and not λk_f .

With $z_{ab}^\chi(X) = \langle J n_{ab} | (X_a^\chi)^{-1} X_b^\chi | n_{ba} \rangle$, the coherent component of the integrand is $\prod_{a < b} (z_{ab}^+)^{2j_{ab}} \overline{(z_{ab}^-)^{2j_{ab}}} = \exp[\lambda S(X) + O(1)]$ with the doubled action

$$S(X) = S_+(X^+) + \overline{S_-(X^-)}, \quad S_\chi = \sum_{a < b} 2k_{ab} \log z_{ab}^\chi. \quad (4.1)$$

The action separates additively in the two chiralities, and since $|z_{ab}^\chi| \leq 1$, $\text{Re } S = \text{Re } S_+ + \text{Re } S_- \leq 0$.

5 The critical set

Proposition 5.1. *Re $S(X) = 0$ if and only if both chiral families satisfy the gluing equations $R(X_a^\chi) n_{ab} = -R(X_b^\chi) n_{ba}$. For nondegenerate Regge-like data each chirality has exactly two solutions up to gauge, related by parity [1]. The zero set of $\text{Re } S$ therefore consists of four symmetry classes. (Not four orbits: each X_a^χ admits an independent central lift $X_a^\chi \mapsto \epsilon_a^\chi X_a^\chi$, $\epsilon_a^\chi = \pm 1$, which leaves $\text{Re } S$ unchanged and, when $\epsilon_a^- = \epsilon_a^+$, leaves every N_a unchanged as well, so each class contains finitely many disconnected lift copies inside the same $\mathcal{D}_\sigma$. That changes a finite multiplicity, not the asymptotic power or phase; the cutoffs in Theorem 7.1 are taken as unions over the lift copies.) The classes are:*

1. *two cross pairings, which reconstruct the two opposite orientations of the nondegenerate simplex, have all $\Omega_a \neq 0$ with a common sign, and lie one in $\mathcal{D}_+$ and one in $\mathcal{D}_-$;*
2. *two same/same pairings, at which $N_a = Y_- Y_+^{-1}$ is independent of a (rank one), so every $\Omega_a = 0$.*

Proof. $\text{Re } S = 0$ forces $|z_{ab}^\chi| = 1$ for every face and chirality, which is saturation of Cauchy–Schwarz, i.e. the gluing equation. Both chiralities carry the same boundary normals, so they have the same two solutions $\{X_a\}, \{X'_a\}$. For the pairing (X, X) , $N_a = Y_- X_a X_a^{-1} Y_+^{-1} = Y_- Y_+^{-1}$ for every a ; five parallel vectors make every 4×4 minor vanish. The cross statement is the reconstruction theorem of [1], and the sign of Ω is that of the reconstructed four-orientation [17]. $\square$

As an independent check, 4000 random starts on the DK gluing equations found exactly two gauge-inequivalent chiral solutions.

Phase at the geometric saddle. At the cross pairing in $\mathcal{D}_\sigma$ the chiral phases are opposite, so $\text{Im } S = 2\sigma \sum_f k_f \Theta_f$ and the oscillatory factor is $\exp[i\sigma 2 \sum_f j_f \Theta_f]$. We keep the area label $A_f = 2j_f$, which is an integer, so the 2π -periodicity argument of Proposition 10.1 applies unchanged. The balanced label $2j_f + 1$ differs by $e^{-i\sigma \sum_f \Theta_f}$; that factor is $O(\lambda^0)$ at fixed geometry but is *not* independent of the bulk once areas are varied, so it may not be absorbed into a boundary convention in Section 10.

6 Sign concentration of the volume operator

Fix a tetrahedron e with classical oriented volume $q_e \neq 0$; take $q_e > 0$, the other case being conjugate, and put $\mu_\lambda = \lambda^3 q_e$. For a state v write $p_{\leq 0}(v) = \|P_{(-\infty, 0]}(Q)v\|^2 / \|v\|^2$. The zero eigenspace is included on purpose: at integer spin Q has a zero eigenvalue, so a bound on $P_{(-\infty, 0)}$ alone would not give $p_+ \rightarrow 1$.

Lemma 6.1 (Ladder estimate). *Let $V_k \subset \bigotimes_f V_{j_f}$ be spanned by vectors obtained from Φ_e by at most k lowering operators, each face in its own frame. Then $(Q - \mu_\lambda)V_k \subset V_{k+3}$ and $\|(Q - \mu_\lambda)|_{V_k}\| \leq C_k \lambda^{5/2}$, with C_k independent of λ .*

Proof. Write $\mathbf{J}_f = (j_f - N_f) \mathbf{n}_f + \mathbf{J}_f^\perp$ in the frame of face f , where N_f counts lowerings and $\mathbf{J}_f^\perp$ is built from $J_f^\pm$. On V_k : $0 \leq N_f \leq k$, $\|J_f^-\| \leq \sqrt{(k+1)2j_f}$ and $\|J_f^+\| \leq \sqrt{k(2j_f+1)}$, both $O(\lambda^{1/2})$. The three faces act on different tensor factors, so the triple product has no ordering ambiguity. Expanding $Q = \mathbf{J}_1 \cdot (\mathbf{J}_2 \times \mathbf{J}_3)$: the term with three longitudinal factors is $\prod_f (j_f - N_f) q = \mu_\lambda + O(\lambda^2)$ on V_k , using $j_f = \lambda k_f + \delta_f$; terms with one transverse factor are $O(\lambda^{5/2})$; with two, $O(\lambda^2)$; with three, $O(\lambda^{3/2})$. Each term raises the lowering count by at most three. $\square$

Lemma 6.2 (Uniform coherent-intertwiner norm). *Let n_f , $f = 1, \dots, 4$, be fixed normals with weights $k_f > 0$ satisfying closure $\sum_f k_f n_f = 0$ and spanning $\mathbb{R}^3$, and let $j_f = \lambda k_f + \delta_f$ be admissible with $|\delta_f| \leq D$. Then there are $c > 0$ and λ_0 , depending only on (n_f, k_f, D) , such that*

$$\|\Pi \Phi_\lambda\|^2 \geq c \lambda^{-3/2} \quad \text{for all } \lambda \geq \lambda_0.$$

Proof. Since Π is an orthogonal projector, $\|\Pi \Phi\|^2 = \langle \Phi | \Pi | \Phi \rangle = \int_{SU(2)} dg \prod_f z_f(g)^{2j_f}$ with $z_f(g) = \langle n_f | g | n_f \rangle$ the spin- $\frac{1}{2}$ overlap, $|z_f| \leq 1$. Write the integrand as $\exp[\lambda S_0(g) + T_\delta(g)]$ with

$$S_0 = \sum_f 2k_f \log z_f, \quad T_\delta = \sum_f 2\delta_f \log z_f.$$

$\text{Re } S_0 \leq 0$, with equality exactly at $g = \pm 1$; at $g = 1$ the first variation is $2i \sum_f k_f n_f = 0$ by closure, and the Hessian of $\text{Re } S_0$ is negative definite because the n_f span, which is the nondegenerate closure saddle of [9]. The perturbation T_δ is holomorphic near $g = 1$ with $T_\delta(1) = 0$, and it and its first two derivatives are bounded by CD uniformly in δ .

Solving $\lambda dS_0 + dT_\delta = 0$ moves the critical point by $u_\delta = O(D/\lambda)$, and the exponent there is $O(D^2/\lambda)$. Ordinary three-dimensional stationary phase with a nondegenerate complex Hessian then gives

$$\int_{|g-1|<\rho} dg e^{\lambda S_0 + T_\delta} = \left(\frac{2\pi}{\lambda}\right)^{3/2} \frac{e^{T_\delta(1)}}{\sqrt{\det(-S_0'')}} (1 + O(D^2/\lambda)),$$

and the complement of the two balls contributes $O(e^{-\eta\lambda})$ by compactness. The point $g = -1$ contributes the same expression times $(-1)^{\sum_f 2j_f}$, which is $+1$ precisely when the spins are

admissible, so the two add rather than cancel. The prefactor depends continuously on δ over the compact set $|\delta_f| \leq D$ and is positive there, so its infimum $c > 0$ is attained, which is the claim. $\square$

Remark 6.3. Two earlier versions of this passage were wrong. The first argued that no uniform constant was needed because the offsets take finitely many values at each λ ; finiteness at each λ gives no single c along an infinite sequence. The second assumed the uniform statement outright. The point that makes the proof work is that the offsets enter only through the bounded holomorphic perturbation T_δ , which shifts the saddle by $O(\lambda^{-1})$ and changes the prefactor continuously; in particular the j -weighted closure defect left by bounded offsets is $O(1)$ against a Hessian of size $O(\lambda)$. The degenerate collinear case, which scales as λ^{-1} , is excluded by the spanning hypothesis.

Lemma 6.4 (Coherent sign concentration). *For each fixed n , $\langle (Q - \mu_\lambda)^{2n} \rangle_{\iota_e} = O(\lambda^{5n+3/2})$, and hence $p_{\leq 0}(\iota_e) = O(\lambda^{-\infty})$.*

Proof. Q commutes with the diagonal $SU(2)$ action, hence with the orthogonal projector Π . Therefore

$$\langle (Q - \mu)^{2n} \rangle_{\iota} = \frac{\|\Pi(Q - \mu)^n \Phi\|^2}{\|\Pi \Phi\|^2} \leq \frac{\|(Q - \mu)^n \Phi\|^2}{\|\Pi \Phi\|^2} \leq \frac{(\prod_{i \leq n} C_{3i})^2 \lambda^{5n}}{\|\Pi \Phi\|^2}$$

by Lemma 6.1, together with the uniform norm bound of Lemma 6.2 below. On the range of $P_{(-\infty, 0]}(Q)$ one has $|Q - \mu| \geq \mu$, so by the spectral theorem $p_{\leq 0} \leq \langle (Q - \mu)^{2n} \rangle / \mu^{2n} = O(\lambda^{3/2-n})$ for every n . $\square$

The only input beyond elementary spin algebra is the norm lower bound, the standard stationary-phase statement for coherent intertwiners. The projection costs a factor $\lambda^{3/2}$, absorbed because n is arbitrary. No uniformity of the constants in n is used or needed: $O(\lambda^{-\infty})$ means that for each N some constant C_N bounds $\lambda^N p_{\leq 0}$.

Reduction to coherent components. The volume weight $\hat{T}_e$ is not a polynomial, so $(\otimes \hat{T}_e)\Psi$ is not a coherent state. Taylor-expand $|q|^{1/2}$ about μ_λ to order M ; the remainder is bounded spectrally by $C_M |q - \mu|^M \mu^{1/2-M}$, and its contribution is $O(\lambda^{3/2-M/2})$ relative to the leading term by Lemma 6.4. The Taylor polynomial applied to ι_e gives finitely many vectors in ΠV_{3M} , whose matrix elements are $e^{\lambda S(X)}$ times polynomials in X of bounded degree. Hence, for every N ,

$$\mathcal{I}_X[(\otimes \hat{T}_e)\Psi] = \sum_{k \leq K_N} a_k(\lambda) P_k(X) e^{\lambda S(X)} + R_N(X), \quad \sup_X |R_N| \leq C_N \lambda^{-N} |a_0(\lambda)|, \quad (6.1)$$

with $a_0 \neq 0$ the leading coefficient.

Lemma 6.5 (Coefficient ratios). *For every fixed $k \geq 1$ and every $\varepsilon > 0$,*

$$\left| \frac{a_k}{a_0} \right| = O(\lambda^{-k/2+k\varepsilon}).$$

In particular $a_1/a_0 = O(\lambda^{-1/2+\varepsilon})$ and every higher term is smaller, so the $k = 0$ term dominates (6.1).

Proof. Put $X = (Q - \mu)/\mu$. Since $\mu \asymp \lambda^3$ and $\|Q\| = O(\lambda^3)$ we have $\|X\| = O(1)$, and Lemma 6.4 gives $\langle |X|^{2n} \rangle = O(\lambda^{3/2-n})$ for every fixed n . Fix $\alpha < 1/2$ and split the spectral measure of X at $|X| = \lambda^{-\alpha}$. On the small part, $\|X^k\| \leq \lambda^{-\alpha k}$. On the large part Markov gives

$$\Pr(|X| > \lambda^{-\alpha}) \leq \lambda^{2\alpha n} \langle |X|^{2n} \rangle = O(\lambda^{3/2-(1-2\alpha)n}),$$

which is smaller than any fixed power once n is chosen large enough, while $\|X\|^k = O(1)$ there. Hence $\|X^k\| = O(\lambda^{-\alpha k})$, and since the k -th Taylor term is $c_k \mu^{1/2} X^k$ against $a_0 = \mu^{1/2}$, the stated bound follows with $\varepsilon = 1/2 - \alpha$. $\square$

Remark 6.6. An earlier version of this lemma bounded the $k = 1$ term by the scalar first moment $\langle Q \rangle - \mu = O(\lambda^2)$. That is invalid: the Taylor term involves the vector $\mu^{-1/2}(Q - \mu)\iota$, and a small expectation value does not bound a norm; used naively Lemma 6.4 would give only $O(\lambda^{1/4})$, which does not even tend to zero. The interpolation above avoids the first moment entirely. We are grateful to the reviewer who supplied it.

7 The domain boundary

The domain $\mathcal{D}_\sigma$ has a boundary, the union of the five cofactor zero loci. Interiority of the geometric saddle validates its local expansion but does not by itself bound the rest of the integral. Review [18] pointed out that a cofactor-order lemma prices certain partly parallel boundary strata at $\lambda^{-7/2}$, ahead of the geometric saddle. That lemma concerns the Barrett–Steele mixed-sign families [4], which exist only when the coherent sign is a free per-face variable. W2 has no such variable.

Theorem 7.1 (Boundary control). *Under Assumption 4.1, the contribution to $\mathcal{A}_\sigma$ from any region of $\partial\mathcal{D}_\sigma$ whose closure stays a positive distance from the rank-one orbits is $O(e^{-\eta\lambda})$ for some $\eta > 0$; deleting those orbits while allowing accumulation would not give a uniform η . The contribution from a neighbourhood of the rank-one orbits is smaller than that of the geometric saddle by a factor $O(\lambda^{-3/2})$. The real-Hessian input (Proposition 8.1) is established for the regular and DK data only, so the second statement carries that restriction.*

Proof. Use (6.1) and a partition of unity $1 = \chi_{\text{geom}} + \chi_{\text{rk1}} + \chi_{\text{rest}}$ on the compact group manifold. On $\text{supp } \chi_{\text{rest}}$, which avoids the four classes of Proposition 5.1 and their central lifts, the continuous function $\text{Re } S$ is strictly negative, hence $\leq -\eta$ by compactness, and every term of (6.1) is $O(\lambda^m e^{-\eta\lambda})$. The geometric orbit in $\mathcal{D}_\sigma$ is interior, since its cofactors are nonzero, so χ_{geom} may be supported inside $\mathcal{D}_\sigma$; the conjugate cross orbit lies in $\mathcal{D}_{-\sigma}$ and does not meet $\overline{\mathcal{D}_\sigma}$.

For the rank-one orbits we bound the modulus of the integrand, which requires no knowledge of the domain shape or of the phase. A rank-one orbit is a pairing (X, X) of one chiral solution with itself, so by (4.1) and Proposition 8.1 the real part of S has a nondegenerate maximum there modulo gauge: $\text{Re } S \leq -c(|u_+|^2 + |u_-|^2)$ in gauge-fixed transverse coordinates $u_\pm$, $c > 0$. All five normals then lie within $O(|u_+| + |u_-|)$ of a common vector, and each cofactor is a 4×4 determinant of four such vectors, so $|\Omega_c| \leq C(|u_+| + |u_-|)^3$. With $D = 24$ the gauge-fixed dimension,

$$\left| \int \chi_{\text{rk1}} \right| \leq C' \int_{\mathbb{R}^D} (|u|)^3 e^{-c\lambda|u|^2} du = O(\lambda^{-D/2-3/2}),$$

while the geometric saddle contributes $c_0 \lambda^{-D/2}$ with $c_0 \neq 0$ by Proposition 8.2 and $|\Omega_c| > 0$ there. The polynomials P_k of (6.1) are bounded on the compact group, and their coefficients obey Lemma 6.5, so the finite sum is dominated by its $k = 0$ term. The ratio is therefore $O(\lambda^{-3/2})$, independently of the radial count of [17], which gives the weaker λ^{-5} against $\lambda^{-9/2}$ and is quoted only as a cross-check from a source that is not yet public. $\square$

The proof uses only Proposition 5.1. As a falsification test we sampled $\sup \text{Re } S$ on the two partly parallel strata named in review, holding the distinct normals at least r apart (Table 1). A positive control at $r = 0$ must recover $\text{Re } S = 0$ at the rank-one corner, and does.

	two-pair, $N_1=N_2, N_3=N_4$	four-parallel, $N_1=\dots=N_4$
control, $r = 0$	-2.0×10^{-10}	-2.4×10^{-11}
$\sup \text{Re } S / \sum_f 2k_f, r \in [0.05, 0.8]$	$-0.040 r^2$	$-0.019 r^2$

Table 1: DK data. The quadratic coefficient is constant to two figures over the whole range; $\text{Re } S$ vanishes on these strata only at the rank-one corner.

8 Nondegeneracy of the geometric saddle

By (4.1) the Hessian of S at any doubled geometric saddle is block diagonal, $\text{diag}(H_+, \overline{H_-})$: the exponent contains no term depending jointly on X^+ and X^- , while $|\Omega_c|$ is an $O(1)$ prefactor. The required nondegeneracy can be established without a numerical determinant.

Proposition 8.1 (Exact real-part coercivity). *At either chiral gluing solution for the regular or the DK simplex, write $X_a = X_a^{(0)} \exp(i \xi_a \cdot \sigma / 2)$ and let m_{ab} be the coherent ray on the edge ab . The quadratic term in the real part of the chiral action is, up to the fixed overall action convention,*

$$-\text{Re } S^{(2)}(\xi) = \frac{1}{4} \sum_{a < b} k_{ab} |(\xi_b - \xi_a) \times m_{ab}|^2.$$

Its kernel is exactly the common $SU(2)$ gauge $\xi_0 = \dots = \xi_4$. Hence after fixing one group element the real Hessian is strictly negative definite.

Proof. For a normalized spinor $|m\rangle$ and a small relative rotation $g = \exp(i d \cdot \sigma / 2)$,

$$|\langle m | g | m \rangle|^2 = 1 - \frac{1}{4} (|d|^2 - (m \cdot d)^2) + O(|d|^3),$$

so each term $2k_{ab} \log \langle m_{ab} | X_a^{-1} X_b | m_{ab} \rangle$ contributes $-\frac{1}{4} k_{ab} |(\xi_b - \xi_a) \times m_{ab}|^2$ to the real quadratic form. Thus a zero mode must satisfy $\xi_b - \xi_a \parallel m_{ab}$ on every edge.

Gauge-fix $\xi_0 = 0$. Exact symbolic facet/bivector arithmetic gives, for both chiralities,

$$\begin{aligned} \text{regular: } \det(b_{01}, b_{02}, b_{12}) &= \pm \frac{25}{95551488}, & |b_{03} \times b_{13}|^2 &= |b_{04} \times b_{14}|^2 = \frac{25}{13759414272}, \\ \text{DK: } \det(b_{01}, b_{02}, b_{12}) &= \pm \frac{1}{5971968}, & |b_{03} \times b_{13}|^2 &= \frac{1}{859963392}, & |b_{04} \times b_{14}|^2 &= \frac{1}{429981696}. \end{aligned}$$

Normalization of the nonzero b_{ab} does not change these independence statements. The first determinant and the edge conditions for 01, 02, 12 force $\xi_1 = \xi_2 = 0$. The two nonparallel-pair identities then force $\xi_3 = 0$ and $\xi_4 = 0$. Therefore the sum of squares has no kernel beyond the common gauge. Since all $k_{ab} > 0$, the gauge-fixed real Hessian is strictly negative definite. The exact certificate is reproduced by `w2_rehessian_kernel_exact.py`; its frozen output is `W2-REHESSIAN-KERNEL-EXACT.log`. $\square$

Proposition 8.2. *On the regular and on the DK simplex, each chiral 15×15 complex Hessian has exactly three null directions, the common $SU(2)$ gauge, and its gauge-fixed 12×12 block is nonsingular. Consequently the full doubled 30×30 Hessian has exactly six $\text{Spin}(4)$ gauge null directions and the gauge-fixed 24×24 Hessian is nonsingular.*

Proof. Write a gauge-fixed chiral Hessian as $H = A + iB$, where $A = \text{Re } H$ and A, B are real symmetric matrices. Proposition 8.1 gives $A < 0$. If $H z = 0$ for a nonzero complex vector z , then

$$0 = \text{Re}(z^\dagger H z) = z^\dagger A z < 0,$$

a contradiction. Thus the gauge-fixed chiral Hessian is invertible. Before gauge fixing, the common left $SU(2)$ action supplies the three and only three null directions. The doubled Hessian is block diagonal in the two chiralities, so the nullities add. $\square$

The earlier numerical calculations are retained as independent checks, not as the proof. Two implementations sharing no Hessian code agree on the regular and DK spectra up to their known normalization factors; finite-difference Richardson extrapolation is stable; and a separately constructed doubled matrix has six gauge nulls. On ten additional random nondegenerate four-simplices the same chiral kernel pattern is observed. Kamiński and Sahlmann’s analytic nondegeneracy theorem for geometric EPRL stationary points is consistent with this result [11], but the present regular/DK proof does not depend on importing their spinor-variable Hessian.

For reference, the numerical conditioning checks give

	regular	DK
$\max \text{ev} / \min_{\neq 0} \text{ev} , 15 \times 15$	1.936	3.854
$\sigma_{\min}, \text{ gauge-fixed } 12 \times 12$	0.1812404	0.0014178 / 0.0013912.

These values are evidence of numerical stability only; nondegeneracy itself is the exact consequence of Proposition 8.1.

9 Main theorem

Theorem 9.1. *Under Assumption 4.1, for the regular or the DK simplex,*

$$\mathcal{A}_\sigma[\Psi] = C_\sigma \lambda^{-p} \exp \left[i\sigma 2 \sum_f j_f \Theta_f + i\phi_0 \right] (1 + O(\lambda^{-1/2+\varepsilon})), \quad C_\sigma \neq 0,$$

where Θ_f are the dihedral angles, ϕ_0 is a fixed boundary-convention phase independent of the bulk, and p is fixed by the 24-dimensional Gaussian and the polynomial normalisations. The conjugate orientation contributes to $\mathcal{A}_{-\sigma}$, not to $\mathcal{A}_\sigma$.

Proof. Lemma 6.4 controls the spectral cusp of $|Q|^{1/2}$: the nonpositive- Q tail of each coherent tetrahedral factor is $O(\lambda^{-\infty})$. By (6.1) the volume-weighted integrand is therefore a finite sum of coherent terms up to $O(\lambda^{-N})$. Theorem 7.1 bounds everything outside a neighbourhood of the geometric saddle by $O(\lambda^{-3/2})$ relative to it. At the saddle, Proposition 8.2 gives a nondegenerate Gaussian and $|\Omega_c| > 0$, so standard stationary phase applies. The leading coefficient is nonzero because the Gaussian is nondegenerate, $|\Omega_c|$ is positive there, and the volume weights are positive on the retained sign sector. The phase is that of Section 5. $\square$

The stated relative error $O(\lambda^{-1/2+\varepsilon})$, for every $\varepsilon > 0$, comes from Lemma 6.5 and needs neither the radial count of [17] nor a symbol expansion. The rank-one boundary term is smaller, $O(\lambda^{-3/2})$ by the absolute-value argument of Theorem 7.1, so it is the $k = 1$ volume-weight term that sets the rate. A sharper $O(\lambda^{-1})$ would require the symbol expansion at the saddle, which is not carried out here. Nothing here uses the finite-spin observation that the wrong-sign weight of the model’s own state is well fitted by $e^{-1.04-0.99j}$ for $j \leq 19$. Its windowed rates drift, and we report the fit as an observation only.

10 Gluing and the area-Regge action

This section is formal calculus of the vertex phase extended to independent area variables. It is not a stationary equation derived for the coherently glued W2 amplitude, for the reason given at the end of the section.

Glue W2 vertices on an oriented simplicial complex with the full intertwiner identity on shared tetrahedra and coherent boundary data compatible across each shared tetrahedron. Orientation compatibility fixes the relative σ_v . By Theorem 9.1, a geometric history carries the phase $\sum_v \sigma_v \sum_{f \subset v} A_f \Theta_{vf}$ with $A_f = 2j_f$.

Proposition 10.1 (Area-Regge stationarity). *With $\delta_f = 2\pi - \sum_{v \ni f} \Theta_{vf}$, the Schläfli identity $\sum_{f \subset v} A_f d\Theta_{vf} = 0$ gives, for internal faces, $dS_{\text{int}} = \sum_f \delta_f dA_f$ up to terms $2\pi A_f$, which change no amplitude because $A_f = 2j_f \in \mathbb{Z}$. The stationary condition in an internal area is $\delta_f \in 2\pi\mathbb{Z}$. In a neighbourhood of a flat background small enough that $|\delta_f| < \pi$ for every internal face, this forces $\delta_f = 0$.*

For the discrete spin sum, Poisson resummation gives the same condition: constructive interference requires $\partial S / \partial A_f \in 2\pi\mathbb{Z}$, and the principal branch is $\delta_f = 0$. This derivation was verified blind against acceptance criteria committed before the derivation was read (Appendix A).

This is the stationary equation of the *area*-Regge action. It is not length Regge, and we do not promote it to length Regge.

Why this is not yet a statement about glued W2. Varying every internal A_f independently is not available on the sector coherent gluing actually retains. Once the shapes on a shared tetrahedron match (Section 11), the areas lie on the length-geometric submanifold $A_f = A_f(l_e)$, and

$$dS = \sum_f \delta_f dA_f = \sum_e \left(\sum_f \delta_f \frac{\partial A_f}{\partial l_e} \right) dl_e,$$

so stationarity gives $\sum_f (\partial A_f / \partial l_e) \delta_f = 0$, the *length*-Regge condition, and not $\delta_f = 0$ face by face. Independent dA_f , and with them the area-Regge equations, require exactly the shape-mismatch directions that coherent gluing suppresses. Proposition 10.1 therefore stands as calculus on independent area variables, not as a derived consequence of the glued amplitude.

11 The Dittrich–Kogios limit

Dittrich and Kogios analyse the area-Regge action on a regular, centrally subdivided hypercubical lattice [5]. They show that its continuum limit reproduces at leading order the graviton dynamics of general relativity, with the first correction at second order in the lattice constant and quadratic in the Weyl curvature, and they note that this holds contrary to expectations from the flatness problem. The condition $\delta_f = 0$ of Proposition 10.1 therefore does not by itself preclude graviton dynamics under refinement.

Their result is obtained from a background-field expansion about flat space on that one lattice family, and they leave the additional suppression terms of effective spin-foam models on that triangulation to future work. We would in any case import it only at that scope, and claim no continuum theorem for a different triangulation, for a nonperturbative model, or for any other implementation of the simplicity constraints. As the next subsection explains, we do not reach even that scope.

The gluing obstruction

An earlier version of this paper asserted a corollary here: that coherently glued W2 histories on the DK lattice carry the area-Regge action, so that the Dittrich–Kogios theorem supplies linearised graviton dynamics. **That inference does not follow, and we withdraw it.** The obstruction was identified in hostile review [19] and is the shape-matching problem in its standard form [6].

Contract two adjacent vertices through the intertwiner identity on a shared tetrahedron. If the asymptotics of both vertices are to be supplied by Theorem 9.1, both sides must present

coherent intertwiners ι_L, ι_R , and the contraction produces their overlap

$$\langle \iota_L | \iota_R \rangle \propto \int dg \prod_f \langle j_f, n_f^L | g | j_f, n_f^R \rangle.$$

Every factor has modulus at most one, and the real exponent vanishes only if a single rotation carries all four normals on one side to those on the other, i.e. only if the two tetrahedral shapes agree. Otherwise compactness gives a strictly negative maximum and the overlap is exponentially suppressed, while the normalisations are only polynomial.

This is exactly the freedom Dittrich and Kogios rely on. Their configuration space contains neighbouring four-simplices inducing *different* shapes on their shared tetrahedron, and the resulting shape-mismatch, area-metric degrees of freedom are integrated out in their derivation. So the chain faces a fork:

- keep the internal data in the coherent regime of Theorem 9.1, and the shared shapes match, so the DK configuration space has *not* been obtained; or
- let the full intertwiner sum explore non-shape-matched data, and Theorem 9.1 supplies no asymptotics for those components.

The obstruction fixes a scale, it does not forbid the modes. The suppression above is stated at *fixed* mismatch. Writing $|\langle \iota_L | \iota_R \rangle| \sim e^{\lambda E(\delta)}$ for a one-parameter shape deformation δ , we find numerically that E is quadratic to leading order, $E(\delta) = -c\delta^2 + O(\delta^3)$ with $c > 0$ fixed by the areas, so at the fluctuation scale $\delta = \epsilon/\sqrt{\lambda}$ the total exponent $\lambda E = -c\epsilon^2$ is *independent of λ* . Measured over $\lambda = 100, 400, 1600$ the value is constant to three figures at each ϵ (`shape_mismatch_scale.py`). Shape mismatches of order $\lambda^{-1/2}$ are therefore not *exponentially* suppressed in λ , which is the scale at which effective spin-foam models carry their area-metric degrees of freedom. They remain finitely suppressed as a function of ϵ , and integrating those variables can still generate powers of $\lambda^{-1/2}$; a single overlap does not fix their net weight in the glued amplitude.

That does not restore the corollary. It says only that the obstruction bounds the *size* of the surviving mismatch rather than excluding it, so the route must be taken at the fluctuation scale rather than at fixed shape mismatch. Closing it still requires deriving the effective glued action after the shared intertwiner contraction, including the measure over the surviving mismatch modes and the stationary-phase prefactor, and showing that its quadratic Hessian on the DK lattice has the required continuum limit. That is not done here, and until it is, this paper proves a vertex theorem and nothing about the continuum. The remaining conditions noted in earlier drafts stand as well: Theorem 9.1 does not control the bulk spin sum, and a history must lie in the support of the state sum.

12 The model's own tetrahedral state

The main theorem uses coherent boundary data. The model also selects its own tetrahedral state, the ground state ψ of H_+ , and it is natural to ask whether Theorem 9.1 holds with ι_e replaced by ψ . This is not implied by Lemma 6.4: ψ is not a coherent intertwiner, and the two states differ measurably (at $j = 8$ the wrong-sign weight is 1.5×10^{-4} for ψ and 1.9×10^{-7} for ι). Two independent reviews found that conflating them is the easiest error to make in this chain.

Exact reduction. Let s_1 be the smallest singular value of Z , so $ZZ^\dagger\psi = s_1^2\psi$. By (2.1),

$$Q\psi = \frac{Z^\dagger Z\psi - s_1^2\psi}{3\sqrt{3}}, \quad \text{Var}_\psi Q = \frac{1}{27} \text{Var}_\psi(Z^\dagger Z).$$

When $2j + 1 \not\equiv 0 \pmod{3}$, $s_1 = 0$ and Chebyshev gives exactly

$$p_{\leq 0}(\psi) \leq \frac{\text{Var}_{\psi} Q}{\langle Q \rangle_{\psi}^2} = \frac{\text{Var}_{\psi}(Z^{\dagger} Z)}{\langle Z^{\dagger} Z \rangle_{\psi}^2}.$$

Conjecture 12.1. *The selected state is an asymptotic eigenvector of $Z^{\dagger} Z$ in relative variance: $\text{Var}_{\psi}(Z^{\dagger} Z) = o(\langle Z^{\dagger} Z \rangle_{\psi}^2)$ as $j \rightarrow \infty$.*

Only $o(1)$ is needed at leading order. Numerically, for $j \leq 20$, the reduction holds to relative precision 10^{-12} , the Chebyshev ratio falls approximately as $1.5 j^{-3}$, $\langle Q \rangle / j^3 \rightarrow 0.80$, and the bound lies above the directly computed wrong-sign weight at every spin. A crude operator-norm estimate loses by j^4 , so a proof must use the structure of ψ .

Uniqueness. For $d = 2j + 1 \leq 41$: $s_1 > 0$ if and only if $d \equiv 0 \pmod{3}$; the smallest singular value is simple at every d , with gap between 1.5 and 182; and the nullity is exactly one when $d \not\equiv 0 \pmod{3}$. The pattern suggests a $\mathbb{Z}_3$ mechanism from the cyclic relabelling $a \rightarrow b \rightarrow c$, under which $Z \mapsto \omega^{\pm 1} Z$. We have not proved it. Since s_1 decays roughly by a factor 0.3 per step of three in d , reaching 3×10^{-6} at $d = 39$, numerics cannot settle uniqueness much beyond $j \approx 25$.

13 Claim boundary

- **Proved.** Theorem 9.1 for coherent boundary data of the regular and DK simplices: the uniform coherent-intertwiner norm is proved in Lemma 6.2, and Propositions 8.2 and 8.1 are certified by the exact real-Hessian kernel argument of Section 8; Proposition 10.1 *only* as formal calculus on independent area variables; it is **not** established for the coherently glued amplitude, where shape matching leaves the length-Regge condition instead (Section 10).
- **Withdrawn.** The continuum inference. Gluing coherent vertices forces matched shapes on shared tetrahedra, which is the freedom the Dittrich–Kogios configuration space requires, so their theorem cannot be applied to glued W2 histories as this paper constructs them (Section 11).
- **Conjectured.** Conjecture 12.1; uniqueness of the selected state for all j .
- **Not claimed.** Lorentzian signature; nonlinear curvature; control of the bulk spin sum of glued histories; dominance for boundary data other than coherent data of nondegenerate simplices (the Hessian is checked explicitly on two); a continuum theorem for generic triangulations; any exponential rate as a theorem; a derivation of general relativity.

14 Relation to existing work

The large-spin analysis of four-simplex amplitudes through coherent states and a doubled chiral action is due to Barrett et al. [1], whose critical-point classification we use directly; the $\text{SU}(2)$ $15j$ case is [2]. Baez, Christensen and Egan found numerical evidence that the Riemannian $10j$ is dominated by degenerate configurations [3], and Barrett and Steele’s stationary-phase analysis accounts for it; the exceptional families that motivated the boundary analysis of Section 7 are Barrett–Steele’s [4]. Coherent intertwiners are Livine–Speziale’s [9]. The appearance of both orientations, as a cosine of the Regge action, and its cure by a projector selecting the Einstein–Hilbert sector are the subject of Engle’s proper vertex [7]. W2 is a close relative, differing in that the amplitude analysed here fixes orientation by the integration domain and uses no additional EH boundary projector. Causal restrictions go back to Livine

and Oriti [8]. The canonical measure is Chaharsough Shirazi and Engle’s [16]. The area-Regge route to gravitons is Dittrich and Kogios’s [5], and area-angle variables are due to Dittrich and Speziale [6]. The algebra of the four-valent volume operator is the Racah algebra [10], and the quantum tetrahedron and the large-spin behaviour of its volume are due to Baez and Barrett [13] and Schliemann [14].

15 Discussion

The vertex does what an orientation-resolved vertex should. On nondegenerate coherent data it has one geometric saddle, with one orientation, and its phase is the area-Regge action. The degenerate configurations that dominate the Barrett–Crane $10j$ are present in W2 as the rank-one orbits, and they lose to the geometric saddle by a power of λ because the canonical measure vanishes cubically on them.

This is a vertex theorem. What stands between it and a derivation of gravity is what stands in front of every spin-foam model: control of the bulk spin sum on glued histories, where the flatness problem lives; nonlinear curvature; and Lorentzian signature. The Dittrich–Kogios result shows that the area-Regge route is not closed by flatness at the linearised level. Whether it survives beyond that is not addressed here.

A Reproducibility and review trail

Frozen objects. The W2 stationary equations, Hessian and sign-concentration argument under review are frozen at commit 818822c841 on branch `sol/w2-epsilon-restriction-audit-2026-09-18`. Independent evidence is on branch `worktree-astra-r2-adjudication-2026-09-19` (PR #350): `w2_hessian.py` and `w2_hessian_dk.py` (Section 8); `nongeometric_criticals.py` (Proposition 5.1); `boundary_strata_res.py` (Table 1); `re_hessian_definite.py` (numerical conditioning cross-checks); `xdx_identity.py`, `b1_moments.py` and `b4_mod3.py` (Section 12); `R3-PREREGISTRATION.md` and `R3-VERIFICATION.md` (Section 10). The exact Hessian certificate is independently frozen on branch `sol/w2-boundary-dominance-2026-09-19` as `w2_rehessian_kernel_exact` (SHA-256 7935701ea3228e4649660ac40e2b724b64feebd3af6a443f06093f6b08fc472c) and `W2-REHESSIAN-KERNEL-EXACT.log` (SHA-256 a806adb55d2eb9e5f87117e69a720e066fde4cbfb4058c39cf8a6).

Reviews that changed this text.

- PR #355, hostile review of S4/S5: established the analytic Hessian, the factor 72 and the six nulls; identified the conflation of coherent and selected states (Section 12); raised the boundary objection answered by Theorem 7.1; required the admissible sequence $j_f = \lambda k_f + O(1)$.
- A context-free review of the sign-concentration argument: the zero-eigenspace correction ($p_{\leq 0}$ rather than p_-), polynomial absorption of the super-polynomial tail against the volume insertions, and the requirement that the moment statement hold for each fixed n .
- A context-free review commissioned on the frozen commit returned UNRESOLVED because it could not obtain the manuscript, and reviewed the brief instead. Two of its findings applied regardless and are incorporated: the Livine–Speziale norm is an asymptotic evaluation rather than a stated uniform inequality, which is now said explicitly in Lemma 6.4 together with the finiteness that makes it sufficient; and the Dittrich–Kogios import must name their background-field expansion and their own deferral of effective spin-foam suppression terms, which Section 11 now does.
- A second context-free review of the repaired manuscript [20] returned ACCEPT_WITH_SCOPE_HARDE and produced the present version. It found the $k = 1$ case of Lemma 6.5 invalid — a

scalar first moment had been substituted for a vector norm — and supplied the interpolation argument that replaces it, which also yields the $O(\lambda^{-1/2+\varepsilon})$ remainder without appealing to the unpublished companion. It found that the continuum withdrawal had not been propagated into Section 10, where independently varied areas are unavailable on the shape-matched sector; that section is now labelled as formal calculus, with the length-Regge condition recorded as what glued W2 actually gives. It hardened the shape-mismatch scale claim from “not suppressed” to “not exponentially suppressed”, sharpened the uniformity discussion in Lemma 6.4 around the $O(1)$ closure defect, required the boundary region in Theorem 7.1 to keep a positive distance from the rank-one orbits, and noted that the evidence bundle claimed a completeness it did not have. It checked and cleared the rank-one absolute-value estimate, the cubic cofactor count, the separation of boundary-state from X -space statements, and the repaired citations.

- The R4 hostile review [21]: placement of the degenerate-dominance condition, which Theorem 7.1 now discharges at the vertex, and the admissibility and area-independence conditions of Section 11.
- A context-free review of the frozen manuscript [19] returned REJECT and is the reason for the present version. It identified the gluing obstruction of Section 11, which removed the continuum corollary and the former subtitle; the $2j_f + 1$ phase defect, repaired by using $A_f = 2j_f$; the insufficiency of the earlier finiteness argument in Lemma 6.4; the unproved coefficient ratios behind the $O(\lambda^{-1})$ remainder; the central sign lifts in Proposition 5.1; the evidentiary status of Propositions 8.2 and 8.1; a spliced bibliography entry for the Racah algebra; and an overstatement of what Baez, Christensen and Egan proved. It also checked and cleared the Taylor remainder of (6.1), the direction of the orthogonal-projection inequality used in Lemma 6.4, and the naming of the numerical conditioning quantities retained in Section 8.
- An earlier hostile-review brief withdrew two of its own claims: that $O(\lambda^{-\infty})$ requires constants uniform in n (false), and that the chiral blocks might couple (excluded by (4.1)).
- A later independent boundary-dominance lane supplied an exact sum-of-squares/kernel certificate for the real chiral Hessian on both the regular and DK backgrounds. Section 8 now uses that certificate to remove the last floating-point nondegeneracy hypothesis: strict negativity of $\text{Re } H$ modulo gauge implies invertibility of the complex gauge-fixed Hessian.

Retracted claims. Five earlier claims of this programme were retracted, including a sign-structure claim (PR #351). The measured $e^{-0.995j}$ law is not claimed.

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- [19] Context-free hostile review of this manuscript at commit `c2d47cfc`, 19 September 2026; disposition REJECT.
- [20] Second context-free hostile review of this manuscript, 19 September 2026; disposition ACCEPT_WITH_SCOPE_HARDENING.
- [21] Hostile review of R4, `orthogonal-validation/astra-r2-adjudication-2026-09-19/R4-HOSTILE-REVI`